

A Disproof of Conjecture 00000000155: There Are More Bell Primes Than B_2 and B_3

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Abstract

Conjecture 00000000155 asserts that the only Bell primes are $B_2 = 2$ and $B_3 = 5$. We refute it: in the same 0-indexed convention used by the conjecture file, $B_7 = 877$ and $B_{13} = 27644437$ are both prime, and neither equals 2 or 5. The claim is therefore false. The computation is formalized in core Lean 4 (`import Std`, no Mathlib) and reproduced with an independent Python script.

1 The Bell numbers

The Bell number B_n counts the partitions of an n -element set, with $B_0 = B_1 = 1$. The conjecture file fixes the convention by writing $B_2 = 2$ and $B_3 = 5$; this is the standard 0-indexed sequence OEIS A000110. The first values, computed with the Bell triangle (Aitken's array),

$$a_{0,0} = 1, \quad a_{n,0} = a_{n-1,n-1}, \quad a_{n,k} = a_{n,k-1} + a_{n-1,k-1},$$

and $B_n = a_{n,0}$, are:

n	B_n
0	1
1	1
2	2 (prime)
3	5 (prime)
4	15
5	52
6	203
7	877 (prime)
8	4140
9	21147
10	115975
11	678570
12	4213597
13	27644437 (prime)

2 Primality certificates

Lemma 1. *877 is prime.*

Proof. Since $\sqrt{877} < 30$, it suffices to rule out the ten primes 2, 3, 5, 7, 11, 13, 17, 19, 23, 29. Direct division gives the nonzero remainders

$$\begin{array}{lll} 877 \bmod 2 = 1, & 877 \bmod 3 = 1, & 877 \bmod 5 = 2, \\ 877 \bmod 7 = 2, & 877 \bmod 11 = 8, & 877 \bmod 13 = 6, \\ 877 \bmod 17 = 10, & 877 \bmod 19 = 3, & 877 \bmod 23 = 3, \\ 877 \bmod 29 = 7. \end{array}$$

No prime $p \leq 29$ divides 877, so 877 is prime. □

Lemma 2. *27644437 is prime.*

Proof. Here $\sqrt{27644437} < 5258$, since $5257^2 = 27636049 \leq 27644437 < 27646564 = 5258^2$. It therefore suffices to check that no odd d with $3 \leq d \leq 5257$ divides 27644437 (the even cases are excluded because the number is odd). An exhaustive trial division, performed exactly in integer arithmetic, finds no divisor; equivalently, the Lean function `isPrimeB` returns `true` on this input, and the Python reproduction script confirms the same. The formal trial-division loop stops as soon as $d^2 > n$, so it examines only $d = 3, 5, \dots, 5257$, a finite check of 2628 candidates. □

3 Refutation

Theorem 3. *The only Bell primes are not $B_2 = 2$ and $B_3 = 5$.*

Proof. By the table, $B_7 = 877$ and $B_{13} = 27644437$, and by the two lemmas both are prime. Since $877 \notin \{2, 5\}$ and $27644437 \notin \{2, 5\}$, the set of Bell primes contains at least $\{2, 5, 877, 27644437\}$, which is strictly larger than $\{2, 5\}$. Hence the conjecture is false. Formally, in `Main.lean` we prove

$$\text{bell } 2 = 2 \wedge \text{bell } 3 = 5 \wedge \text{isPrimeB } (\text{bell } 7) = \text{true} \wedge \text{isPrimeB } (\text{bell } 13) = \text{true},$$

together with `bell 7 ≠ bell 2` and `bell 7 ≠ bell 3`, all by kernel reduction with `decide` (no `sorry`, no `axiom`, no `native_decide`). □

Conjecture 4 (00000000155, refuted). The only Bell primes are $B_2 = 2$ and $B_3 = 5$.

Corollary 5. *The list of Bell primes begins 2, 5, 877, 27644437, ...; in particular the conjecture is incomplete. We verify the range $0 \leq n \leq 20$ here; larger Bell primes are reported in the literature, but are not needed for the refutation.*